

NUnit and Moq

1. Write Testable Code with Moq :

Task1

In this task, you will create a class library that will be used for unit testing.

CustomerComm.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace CustomerCommLib
{
    public class CustomerComm
    {
        private IMailSender _mailSender;

        public CustomerComm(IMailSender mailSender)
        {
            _mailSender = mailSender;
        }

        public bool SendMailToCustomer()
        {
            string customerEmail = "cust123@abc.com";
            string message = "Some Message";
```

```

        _mailSender.SendMail(customerEmail, message);

        return true;
    }
}

```

MailSender.cs :

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Net;
using System.Net.Mail;

namespace CustomerCommLib
{
    public class MailSender : IMailSender
    {
        public bool SendMail(string toAddress, string message)
        {
            MailMessage mail = new MailMessage();
            SmtpClient smtpServer = new SmtpClient("smtp.gmail.com");

            mail.From = new MailAddress("your_email_address@gmail.com");
            mail.To.Add(toAddress);
            mail.Subject = "Test Mail";
            mail.Body = message;

```

```

smtpServer.Port = 587;
smtpServer.Credentials = new NetworkCredential("username", "password");
smtpServer.EnableSsl = true;

smtpServer.Send(mail);

return true;
}
}
}

```

IMailSender.cs :

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace CustomerCommLib
{
    public interface IMailSender
    {
        bool SendMail(string toAddress, string message);
    }
}

```

Task2

In this task, you will create unit test project which make use of NUnit framework and Moq.

CustomerCommTests.cs :

```
using System;
using System.Collections.Generic;
using System.Text;
using NUnit.Framework;
using Moq;
using CustomerCommLib;

namespace CustomerComm.Tests
{
    [TestFixture]
    public class CustomerCommTests
    {
        private Mock<IMailSender> _mailSenderMock = null!;
        private CustomerCommLib.CustomerComm _customerComm = null!;

        [OneTimeSetUp]
        public void Setup()
        {
            _mailSenderMock = new Mock<IMailSender>();

            _mailSenderMock
                .Setup(x => x.SendMail(
                    It.IsAny<string>(),
                    It.IsAny<string>()))
```

```

        .Returns(true);

        _customerComm =
            _customerComm
            = new
CustomerCommLib.CustomerComm(_mailSenderMock.Object);
    }

[Test]
[TestCase]
public void SendMailToCustomer_ShouldReturnTrue()
{
    bool result =
        _customerComm.SendMailToCustomer();

    Assert.IsTrue(result);
}
}
}

```

Output :

